Diffuse acoustic waves in a randomly stratified flow

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Abstract

In this communication we develop an integral representation of the acoustic waves emitted by a source and transmitted by a randomly stratified fluid flow. The analysis is carried out in a regime of separation of scales whereby the fluctuations of the flow are much smaller than the source wavelength, which in turn is much smaller than the thickness of the flow–the so-called diffusion regime. Our aim is to subsequently develop coherent interferometric (CINT) imaging algorithms based on cross-correlation functions of the acoustic waves recorded at the bottom of the flow to possibly locate the source above it.

Keywords: Aero-acoustics, Diffusion, Coherent interferometry.

Introduction

The purpose of imaging techniques is to estimate the location of one or more sources and/or reflecting structures with a passive or an active array of receivers. Coherent interferometric (CINT) algorithms [1] have been developed for imaging in cluttered media from the time traces of echoes recorded at a remote array. The main effect of the clutter is to induce large delay spread, or coda, to the recorded time traces in regimes where significant multiple scattering of the acoustic waves occurs. In this work we are more particularly interested in the propagation of acoustic waves in a stratified, heterogeneous flow and the possible localization of sources by these wave fields. CINT imaging is based on the back-propagation of local space-time empirical cross-correlations of the array data, namely the recorded pressure fields at the near free-surface of an half-space or a random slab [3,4]. Our ultimate objective here is to extend CINT imaging techniques to possibly account for the influence of a convecting shear flow.

For that purpose, we study here the acoustic waves emitted by a source and transmitted by a randomly stratified fluid flow. We start from the earlier works of Garnier *et al.* [3,4] and Borcea *et al.* [2] and consider successively the influence of an homogeneous background flow, and a randomly stratified background flow. The analysis is based on an assumption of separation of scales, whereby the fluctuations of the flow are much smaller than the source wavelength, which in turn is much smaller than the thickness of the flow.

Acoustic waves in an homogeneous flow

We consider the Euler equations for a compressible fluid flow and linearize them about an unperturbed, stationary flow for which the pressure, fluid velocity, and fluid density do not depend on time t. They are denoted by $p_0(\boldsymbol{r})$, $\boldsymbol{v}_0(\boldsymbol{r})$ and $\varrho_0(\boldsymbol{r})$ respectively, where $\boldsymbol{r} \in \mathbb{R}^3$ stands for the position, such that:

$$\begin{aligned}(\boldsymbol{v}_0 \cdot \boldsymbol{\nabla})\varrho_0 &= -\varrho_0 \boldsymbol{\nabla} \cdot \boldsymbol{v}_0 \,,\\ (\boldsymbol{v}_0 \cdot \boldsymbol{\nabla})\boldsymbol{v}_0 &= -\frac{1}{\varrho_0}\boldsymbol{\nabla} p_0 \,,\\ (\boldsymbol{v}_0 \cdot \boldsymbol{\nabla})p_0 &= c_0^2(\boldsymbol{v}_0 \cdot \boldsymbol{\nabla})\varrho_0 \,.\end{aligned}$$

Here c_0 stands for the sound velocity not influenced by the waves. Linearization consists in assuming that the actual flow is a perturbation $(\varrho', \boldsymbol{v}', p')$ of the stationary flow generated by a source $\boldsymbol{f}(t,\boldsymbol{r})$ per unit mass (neglecting gravity):

$$\delta(t,\boldsymbol{r}) = \delta_0(\boldsymbol{r}) + \delta'(t,\boldsymbol{r}) \,,$$

where $\delta \in \{\varrho, \boldsymbol{v}, p\}$. The primed quantities p', $\boldsymbol{v}'$ and ϱ' are the acoustic pressure, fluid velocity and density, respectively, of which non-linear contributions to the Euler equations above are assumed negligible. We note $\boldsymbol{r} = (\boldsymbol{x}, z) \in \mathbb{R}^2 \times \mathbb{R}$ where $\boldsymbol{x}$ stands for the horizontal coordinates of the position, and z stands for the vertical coordinate. We start with the case where the unperturbed background flow is homogeneous, that is $\boldsymbol{v}_0$ and ϱ_0 are non vanishing and constant in a slab $z \in [-L, 0]$. In addition the background flow celerity $\boldsymbol{v}_0$ is horizontal, a situation which arises in many instances of real media (oceans, Earth's crust, atmosphere, jet flows, *etc*). Then the acoustic velocity $\boldsymbol{v}'$ and pressure p' satisfy:

$$(\partial_t + \boldsymbol{v}_0 \cdot \boldsymbol{\nabla}_{\boldsymbol{x}})p' + K_0 \boldsymbol{\nabla} \cdot \boldsymbol{v}' = 0\,,$$
$$(\partial_t + \boldsymbol{v}_0 \cdot \boldsymbol{\nabla}_{\boldsymbol{x}})\boldsymbol{v}' + \frac{1}{\varrho_0}\boldsymbol{\nabla} p' = \boldsymbol{f}\,,$$

where $K_0 = \varrho_0 c_0^2$ is the fluid compressibility, and $\boldsymbol{\nabla}_{\boldsymbol{x}}$ stands for the gradient in the horizontal plane. With the appropriate Fourier transform (FT) with respect to time t and horizontal coordinates $\boldsymbol{x}$, we can solve this system for a source $\boldsymbol{f}(t,\boldsymbol{r}) = \boldsymbol{F}(t,\boldsymbol{x})\delta(z - z_s)$ with $z_s \geq 0$ and show that the acoustic pressure below the slab has the integral representation:

$$p'(t,\boldsymbol{x},-L) = \frac{1}{16\pi^3}\iint \mathrm{e}^{-\mathrm{i}\omega(t-\boldsymbol{\kappa}\cdot\boldsymbol{x}-L\zeta(\boldsymbol{\kappa}))}$$
$$\times \Big(\frac{\varrho_0 \boldsymbol{\kappa}\cdot\widehat{\boldsymbol{F}}_{\boldsymbol{x}}(\omega,\boldsymbol{\kappa})}{\beta(\boldsymbol{\kappa})\zeta(\boldsymbol{\kappa})} - \varrho_0 \widehat{F}_z(\omega,\boldsymbol{\kappa})\Big)\omega^2\, d\omega d\boldsymbol{\kappa}\,,$$

where $\zeta(\boldsymbol{\kappa}) = (\frac{\beta}{c_0^2} - \frac{\kappa^2}{\beta})^{\frac{1}{2}}$, $\beta = 1 - \boldsymbol{v}_0 \cdot \boldsymbol{\kappa}$, and $(\widehat{\boldsymbol{F}}_{\boldsymbol{x}}(\omega,\boldsymbol{\kappa}), \widehat{F}_z(\omega,\boldsymbol{\kappa}))$ is the FT of the three-dimensional source $\boldsymbol{F}$, ω is the circular frequency, and $\boldsymbol{\kappa}$ is the horizontal slowness vector.

Acoustic waves in a random flow

We now consider the case where the bulk modulus K_0 is randomly varying about an homogenized value $\underline{K} = \varrho_0 \underline{c}^2$ and address the diffusion approximation regime [3], whereby the scale of fluctuations ℓ of the former is much smaller than the typical wavelength λ of the waves, which in turn is much smaller than the depth of the flow L. This scaling is quantified by introducing the small parameter $0 < \varepsilon \ll 1$ such that $\frac{\ell}{\lambda} = \varepsilon$ and $\frac{\ell}{L} = \varepsilon^2$. In this respect, the bulk modulus depends on the depth z in the flow $(-L, 0)$ and is constant outside:

$$\frac{1}{K_0(z)} = \begin{cases} \frac{1}{\underline{K}}\left[1 + \nu\left(\frac{z}{\varepsilon^2}\right)\right] & \text{for } z \in [-L, 0]\,, \\ \frac{1}{\underline{K}} & \text{elsewhere}\,, \end{cases}$$

where $(\nu(z))_{z\in\mathbb{R}}$ is a zero-mean, second-order stochastic process. Also the source is assumed to vary at the scale ε and is denoted by $\boldsymbol{F}^\varepsilon(t,\boldsymbol{x}) = \boldsymbol{F}(t/\varepsilon, \boldsymbol{x}/\varepsilon)$. Then the acoustic pressure admits the following integral representation:

$$p'_\varepsilon(t,\boldsymbol{x},-L) = \frac{1}{16\pi^3}\iint \mathrm{e}^{-\frac{\mathrm{i}\omega}{\varepsilon}(t-\boldsymbol{\kappa}\cdot\boldsymbol{x}-L\underline{\zeta}(\boldsymbol{\kappa}))} T^\varepsilon(\omega,\boldsymbol{\kappa})$$
$$\times \Big(\frac{\varrho_0 \boldsymbol{\kappa}\cdot\widehat{\boldsymbol{F}}^\varepsilon_{\boldsymbol{x}}(\omega,\boldsymbol{\kappa})}{\beta(\boldsymbol{\kappa})\underline{\zeta}(\boldsymbol{\kappa})} - \varrho_0 \widehat{F}^\varepsilon_z(\omega,\boldsymbol{\kappa})\Big)\omega^2\, d\omega d\boldsymbol{\kappa}\,,$$

where $\underline{\zeta}(\boldsymbol{\kappa}) = (\frac{\beta}{\underline{c}^2} - \frac{\kappa^2}{\beta})^{\frac{1}{2}}$, and $T^\varepsilon(\omega,\boldsymbol{\kappa})$ is the transmission coefficient which defines how the acoustic waves cross the slab. Its properties are fully characterized as in [3]. Essentialy the main difference between the present situation and the situation addressed in [3] where no background flow is considered lies in the phase ϕ. In the latter situation it is $\phi(t,\boldsymbol{x},\boldsymbol{\kappa}) = t - \boldsymbol{\kappa}\cdot\boldsymbol{x} - \frac{L}{c(\boldsymbol{\kappa})}$ with $c(\boldsymbol{\kappa}) = \frac{\underline{c}}{\sqrt{1-\kappa^2\underline{c}^2}}$, whereas in the situation the celerity of the background flow has an important influence, as expected: $\phi(t,\boldsymbol{x},\boldsymbol{\kappa}) = t - \boldsymbol{\kappa}\cdot\boldsymbol{x} - L\underline{\zeta}(\boldsymbol{\kappa})$, with the previous definition of $\underline{\zeta}$.

In future works we shall analyze how this result compares with the existing experiments and numerical simulations reported in the literature [5, 6], and develop a CINT imaging functional of the source $\boldsymbol{f}$ based on the integral representation established above.

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